

Chapter 119 — Genitourinary Complications of Mesh and Their Repair

- Mesh is used for **male and female stress urinary incontinence (SUI)** and **female pelvic organ prolapse (POP)**.
- Its use is **among the most controversial subjects in female pelvic medicine and reconstructive surgery**, the dispute turning on whether the harms of mesh outweigh its benefits.
- Harms fall into three groups: **mesh-related pain**; **exposure, extrusion or erosion**; and **damage to adjacent structures** — nerves, muscles, organs, vasculature — at implantation.
- The surgical obligation is twofold: **selection and counselling before implantation**, and **diagnosis and treatment of complications** afterwards.

MESH AND ITS PROPERTIES

Why the pelvic floor is not the abdominal wall

- Polypropylene has been in surgical use **over 50 years**; its success against hernia recurrence drove its migration into other compartments.
- The pelvic floor differs from the abdominal wall in three ways that predispose to complications: **proximity to bladder, urethra, vagina, perineum and rectum**; **substantial physiological stretch**; **abundant local microbiota**.
- Adverse outcomes: **infection, chronic pain, dyspareunia, exposure, shrinkage, perforation into surrounding structures, lower urinary tract symptoms (LUTS)**.
- Advantages over native tissue, cadaveric tissue and xenograft: **no morbid harvest, no human or animal material and therefore no disease transmission**, moderate cost, versatility, durability.

Classification by pore size

MESH TYPES I–IV BY PORE SIZE (AMID CLASSIFICATION, 1997)

Type	Pore size	Examples	Behaviour in the vagina
I	Macroporous, >75 µm	Polypropylene — knitted, and woven (Marlex)	Fibrocollagenous ingrowth and integration; the only recommended implantable mesh
II	Microporous; expanded polytetrafluoroethylene ≈10 µm	Polytetrafluoroethylene (PTFE, Teflon); Gore-Tex (expanded PTFE)	Encapsulated rather than integrated; full-length slings frequently required excision
III	Macropores or perforations set in a microporous base	Mycromesh; Mersilene (knitted polyethylene terephthalate)	Extrusion requiring excision, and voiding symptoms
IV	Submicronic	Silicone grafts and silicone-coated mesh	Vaginal extrusion and sinus formation, commonly

- The classification is **single-variable**: pore size alone assigns the type, and integration, infection risk and extrusion follow from it.
- Within type I the contrast is **configuration, not polymer**: **woven polypropylene (Marlex) is among the stiffest materials**, wrinkles and extrudes readily, whereas **knitted polypropylene is softer and more compliant**, its loose knit giving larger interstices (**≈1500 µm against ≈600 µm**) and greater elasticity.
- A **melting point of 160°C** permits autoclaving.
- Mersilene derives **bidirectional elasticity from interlinks at each fibre junction** yet still extrudes: **elasticity does not protect against extrusion**.

Pore size and wound healing

- The governing principle is a **size mismatch between bacteria and host immune cells**: bacteria $\approx 1\ \mu\text{m}$, macrophages and lymphocytes $\approx 50\ \mu\text{m}$.
- **Small pores therefore admit bacteria while excluding macrophages and lymphocytes**, seeding infection where the immune system cannot reach it.
- **Large pores** — type I polypropylene, $\approx 1500\ \mu\text{m}$ — **permit fibrocollagenous ingrowth**, integrating the mesh into local tissue and conferring strength and support.
- Attempts to improve tissue acceptance by **surface modification rather than pore geometry have failed**: bovine collagen injection (ProteGen) and novel welding (Uratape/ObTape) both produced **unacceptable rates of mesh exposure and fistula formation**.

Long-term biological consequences

- **The cancer hypothesis is disproven**. Chronic foreign-body response was held to raise malignancy risk; population-based follow-up shows **no excess of pelvic or any other cancer**.
- **No association with autoimmune disease** on long-term follow-up.
- Histopathology after vaginal mesh placement: **moderate vaginal fibrosis in more than half** and **giant cell reaction in almost all**, mild inflammation predominating overall.
- Counterintuitively, **extrusion or pain is associated with less inflammation than voiding dysfunction** — inflammatory intensity does not track the presenting symptom.

Concept — pore size is the single variable from which all mesh behaviour is generated: it decides whether host immune cells can follow bacteria into the mesh, and whether fibrocollagenous tissue can grow through it. Every complication profile in the Amid classification is downstream of that one dimension, and every attempt to buy biocompatibility through surface chemistry rather than pore geometry has failed.

TERMINOLOGY AND FEDERAL REGULATIONS OF GENITOURINARY MESH

- Terminology has been inconsistent over time. The **joint International Urogynecological Association (IUGA) and International Continence Society (ICS) definitions** are the standard used throughout the chapter, an interactive version being hosted by the ICS.

IUGA/ICS TERMINOLOGY FOR MESH COMPLICATIONS

Term	Definition	Status
Erosion	Tissue worn away by friction or pressure	Deprecated — do not use of mesh
Exposure	Mesh visible through separated or non-healing vaginal epithelium	Correct term
Extrusion	Gradual passage out of a body structure or tissue — a loop of sling protruding into vagina or bladder, not via the surgical incision site	Correct term
Perforation	Abnormal opening into a hollow organ or viscus	Correct term
Contraction	Shrinkage or reduction in size of the mesh	Correct term
Prominence	Definition pending — not yet available in the source material to hand	Correct term

- The distinction the table enforces is **route and visibility, not severity**: exposure is **seen** through unhealed epithelium, extrusion has **travelled out** by a route other than the incision, perforation has **entered a viscus**. "Erosion" is rejected precisely because it identifies none of these.